

A Negative Answer to the Grothendieck Question for $c_{0,\mathcal{W}\mathcal{O}}$

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Status

Candidate counterexample / negative answer, likely valid. This packet answers Question 5.23 of Rincon-Villamizar and Uzcategui Aylwin [1]: $c_{0,\mathcal{W}\mathcal{O}}$ is not a Grothendieck space.

Source Question

The source paper studies the Banach spaces

$$c_{0,\mathcal{I}} = \{x \in \ell_\infty : \mathcal{I}\text{-}\lim x_n = 0\}$$

associated with ideals $\mathcal{I}$ on countable sets. Near the end of Section 5, after several positive and negative results about the Grothendieck property, the authors ask:

Theorem 5.22. *If $\mathcal{I}$ is an intersection of finitely many maximal ideals on $\mathbb{N}$, then $c_{0,\mathcal{I}^\omega}$ is complemented in ℓ_∞ and, in particular, is Grothendieck.*

Proof. From the hypothesis, we know that $c_{0,\mathcal{I}}$ is complemented in ℓ_∞ (Proposition 5.21). Let $Y \subset \ell_\infty$ be a subspace such that $\ell_\infty = c_{0,\mathcal{I}} \oplus Y$. So,

$$\ell_\infty \sim \ell_\infty(\ell_\infty) \sim \ell_\infty(c_{0,\mathcal{I}}) \oplus_\infty \ell_\infty(Y) \sim c_{0,\mathcal{I}^\omega} \oplus_\infty \ell_\infty(Y),$$

by Theorem 5.2. Whence $c_{0,\mathcal{I}^\omega}$ is isomorphic to a complemented subspace of ℓ_∞ . $\square$

Question 5.23. Is $c_{0,\mathcal{W}\mathcal{O}}$ a Grothendieck space?

ACKNOWLEDGMENTS

Part of the research of this paper was developed during a postdoctoral stay of the first au

In text form, Question 5.23 asks:

Is $c_{0,\mathcal{W}\mathcal{O}}$ a Grothendieck space?

Definitions and Source Facts

If $\mathcal{I}$ is an ideal on a countable set S , then

$$c_{0,\mathcal{I}} = \{x \in \ell_\infty(S) : \{s \in S : |x_s| \geq \varepsilon\} \in \mathcal{I} \text{ for every } \varepsilon > 0\}.$$

For $A \subseteq S$, the restriction ideal is

$$\mathcal{I}|A = \{A \cap B : B \in \mathcal{I}\}.$$

We use the following facts from [1].

- (1) Proposition 3.8 states that $c_{0,\mathcal{I}|A}$ is Banach-lattice isometric to a closed ideal of $c_{0,\mathcal{I}}$.
- (2) Theorem 5.13 states that every ideal in the family $\mathcal{B}$ is isomorphic to a restriction $\text{WO}|A$ for some $A \subseteq \mathbb{Q}$.
- (3) The family $\mathcal{B}$ contains $P_1 = \text{Fin}^\omega$.
- (4) Theorem 5.19 states that c_{0,P_α} and c_{0,Q_α} are not Grothendieck for every $1 \leq \alpha < \omega_1$. In particular, c_{0,P_1} is not Grothendieck.
- (5) The source recalls in Section 5.3 that the Grothendieck property passes to complemented subspaces.

Proof Intuition

The source paper already places many non-Grothendieck spaces c_{0,P_α} and c_{0,Q_α} inside $c_{0,\text{WO}}$ as closed subspaces. A closed subspace alone does not obstruct the Grothendieck property, since Grothendieck spaces may contain non-Grothendieck subspaces. The missing point is that restriction subspaces are coordinate subspaces, and coordinate projection onto the restriction is contractive. Thus the copy of c_{0,P_1} obtained from the restriction theorem is actually complemented in $c_{0,\text{WO}}$. Since the Grothendieck property is inherited by complemented subspaces, this complemented copy rules out Grothendieckness of $c_{0,\text{WO}}$.

Complemented Restriction Lemma

Lemma 1. *Let $\mathcal{I}$ be an ideal on a countable set S , and let $A \subseteq S$. The restriction copy of $c_{0,\mathcal{I}|A}$ inside $c_{0,\mathcal{I}}$ is the range of a norm-one lattice projection. In particular, it is 1-complemented in $c_{0,\mathcal{I}}$.*

Proof. Identify $c_{0,\mathcal{I}|A}$ with

$$E_A = \{x \in c_{0,\mathcal{I}} : x_s = 0 \text{ for every } s \notin A\}.$$

Define $P_A : c_{0,\mathcal{I}} \rightarrow \ell_\infty(S)$ by

$$(P_A x)_s = \begin{cases} x_s, & s \in A, \\ 0, & s \notin A. \end{cases}$$

If $x \in c_{0,\mathcal{I}}$ and $\varepsilon > 0$, then

$$\{s \in S : |(P_A x)_s| \geq \varepsilon\} = A \cap \{s \in S : |x_s| \geq \varepsilon\}.$$

The second set belongs to $\mathcal{I}$, and ideals are hereditary; hence the intersection belongs to $\mathcal{I}$. Therefore $P_A x \in c_{0,\mathcal{I}}$. Moreover $P_A^2 = P_A$, $\|P_A\| \leq 1$, and P_A fixes precisely the subspace E_A . Thus E_A is 1-complemented. Under restriction to the coordinates in A , E_A is Banach-lattice isometric to $c_{0,\mathcal{I}|A}$. $\square$

Main Result

Theorem 1. *The Banach space $c_{0,\text{WO}}$ is not Grothendieck.*

Proof. By [1, Theorem 5.13], applied to $P_1 \in \mathcal{B}$, there is a set $A \subseteq \mathbb{Q}$ such that P_1 is isomorphic to the restriction ideal $\mathbf{WO}|_A$. By the Banach-lattice Banach–Stone result for these spaces [1, Theorem 3.3], this gives a Banach-lattice isometry

$$c_{0,P_1} \cong c_{0,\mathbf{WO}|_A}.$$

By the complemented restriction lemma, $c_{0,\mathbf{WO}|_A}$ is 1-complemented in $c_{0,\mathbf{WO}}$. Hence $c_{0,\mathbf{WO}}$ contains a complemented subspace isometric to c_{0,P_1} .

The source paper proves that c_{0,P_1} is not Grothendieck [1, Theorem 5.19]. If $c_{0,\mathbf{WO}}$ were Grothendieck, then every complemented subspace of $c_{0,\mathbf{WO}}$ would be Grothendieck, contradicting the complemented copy of c_{0,P_1} . Therefore $c_{0,\mathbf{WO}}$ is not a Grothendieck space. $\square$

Verification Notes

No computational verification is involved. The proof is a short permanence argument. The only new step beyond the source paper is the explicit coordinate projection P_A , whose range is the restriction copy.

A reviewer should check:

- that the source’s theorem representing P_1 as a restriction $\mathbf{WO}|_A$ is being applied with the same notion of ideal isomorphism used by the Banach-lattice isometry theorem;
- that the coordinate projection P_A maps $c_{0,\mathbf{WO}}$ into itself;
- that the Grothendieck property is used only through its standard permanence for complemented subspaces.

Novelty and Literature Check

Local run indexes searched before promotion:

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`.

The search terms included 2309.08076, $c_{\{0,\mathbf{WO}\}}$, $\mathbf{WO}$, Grothendieck, and I-convergent sequences. No prior packet or attempt for this target was found.

Web searches on 2026-06-29 for the exact and close phrases

" $c_{\{0,\mathbf{WO}\}}$ " Grothendieck, " $c_{0,\mathbf{WO}}$ " Grothendieck space, "Is $c_{\{0,\mathbf{WO}\}}$ " "Grothendieck", and the source title together with $\mathbf{WO}$ found only the original arXiv record [1], with no later explicit answer located in this bounded pass.

Limitations

This packet gives a negative answer only for the specific final question about $c_{0,\mathbf{WO}}$. It does not classify which ideals $\mathcal{I}$ make $c_{0,\mathcal{I}}$ Grothendieck.

Human Review Recommendation

Likely valid. The proof is short and uses the source paper’s own structural results; review should focus on the restriction-complementation observation.

References

- [1] M. A. Rincon-Villamizar and C. Uzcategui Aylwin, *Banach spaces of $\mathcal{I}$ -convergent sequences*, arXiv:2309.08076, 2023.
- [2] M. Gonzalez and T. Kania, *Grothendieck spaces: the landscape and perspectives*, Japan. J. Math. 16 (2021), no. 2, 247–313.